

Non-Contact Heartbeat Detection with mmWave Radar Using Impulse Trains

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Abstract—This paper presents a non-contact heartbeat detection method using a mmWave radar system. The approach integrates range FFT, high-resolution angle estimation, and respiration-guided target selection to extract weak chest micro-motions. A heartbeat estimation algorithm based on impulse train modeling is then applied to improve robustness against noise and respiratory interference. Experiments with four human subjects at varying distances and angles show that the proposed method outperforms traditional FFT and autocorrelation (ACF), achieving an average detection error of 3.29%, compared to 9.11% (FFT) and 7.80% (ACF). These results confirm the effectiveness of mmWave radar for unobtrusive and reliable indoor vital sign monitoring.

Index Terms—mmWave radar, heartbeat detection, non-contact sensing, frequency-domain analysis, impulse train

I. INTRODUCTION

With the rise of smart healthcare, sleep monitoring, and health management, non-contact vital sign monitoring has attracted growing attention. Compared to contact-based methods like ECG and PPG, non-contact approaches offer greater comfort and privacy. Among them, millimeter-wave (mmWave) radar stands out for its non-invasive nature, high resolution, and sensitivity to subtle chest motions from respiration and heartbeat. It also performs reliably under low light and partial occlusion, making it well-suited for practical use.

Research on mmWave-based monitoring has focused on two key aspects: target localization and periodic signal detection. For the former, common methods include CFAR detection and peak extraction from range-Doppler maps [1], or multi-frame energy accumulation for improved stability

[2]. For the latter, phase demodulation followed by bandpass filtering is often used to separate respiration and heartbeat components [3], with heartbeat estimated via autocorrelation, FFT, or envelope analysis [4]. Advanced techniques such as EMD [5] and wavelet transform [6] have also been explored.

However, many existing methods lack effective mechanisms to distinguish human targets from static reflectors or non-living objects, often relying on simple amplitude-based heuristics, which may include irrelevant or low-quality signals. Moreover, heartbeat estimation often relies on local peak detection in the phase signal, which is highly susceptible to respiratory harmonics and noise, making it difficult to isolate the heartbeat component—especially under low-SNR or multi-target conditions [7].

To address these challenges, we propose a robust multi-target framework based on high-resolution spectral estimation, respiration-guided target evaluation, and impulse-train-based heartbeat detection.

The remainder of this paper is organized as follows. Section II details the proposed method, Section III presents experimental results, and Section IV concludes the paper.

II. METHODOLOGY

The proposed system comprises radar signal acquisition, range processing, target selection, phase-based signal extraction, and frequency analysis. As shown in Fig. 1, it processes raw IF signals from the mmWave radar to estimate heartbeat non-contactly.

Specifically, raw ADC data is collected using an AWR1642 radar and DCA1000 board. A range FFT identifies active bins, followed by high-resolution angle estimation via MUSIC. Candidate points are selected through non-maximum suppression and evaluated by their respiration power ratio. Heartbeat detection is then performed using CFAR-based spike detection, and the frequency is estimated by applying FFT to a constructed impulse train.

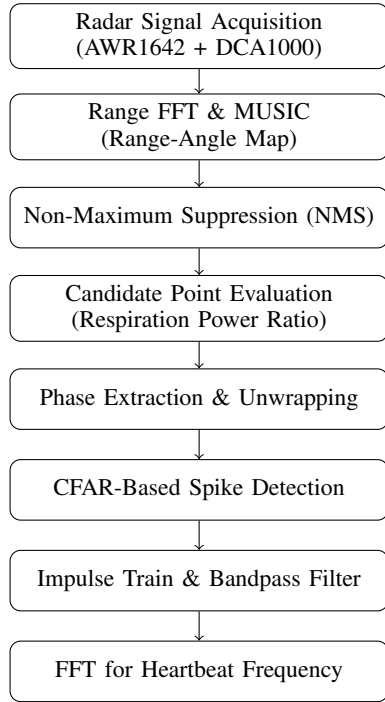


Fig. 1: System pipeline for heartbeat detection based on mmWave radar.

A. Human Target Extraction and Respiration Rate Detection

This section outlines the procedure for identifying human targets from raw mmWave radar signals using spectral estimation and respiration-based evaluation. First, a range-angle map is generated by applying range FFT and high-resolution angle estimation (e.g., MUSIC). Non-maximum suppression (NMS) is then applied to identify candidate target points. For each candidate, the slow-time phase signal is extracted and its respiration spectrum is computed. The power concentration

within the expected breathing frequency band is used to calculate a respiration power ratio, and the candidates with the highest ratios are selected as likely human targets.

1) *Range FFT*: We first apply a one-dimensional FFT along the fast-time axis to convert the time-domain signal $s(n, m, k)$ into the range domain, where n is the fast-time index, m the frame index, and k the antenna index. For each frame m and antenna k , the range profile is:

$$S(r, m, k) = \sum_{n=0}^{N-1} s(n, m, k) \cdot e^{-j2\pi \frac{nr}{N}}, \quad (1)$$

yielding a 3D tensor $S(r, m, k)$ over range, time, and antenna.

2) *MUSIC-Based Angle Estimation*: For each range bin r , we construct the antenna snapshot covariance matrix:

$$\mathbf{R}_r = \frac{1}{M} \sum_{m=1}^M \mathbf{s}_r(m) \mathbf{s}_r(m)^H, \quad (2)$$

where $\mathbf{s}_r(m)$ is the k -dimensional vector at range r and frame m . We then compute the AoA spectrum via the MUSIC algorithm [8]:

$$P_{\text{MUSIC}}(\theta, r) = \frac{1}{\mathbf{a}^H(\theta) \mathbf{E}_{n,r} \mathbf{E}_{n,r}^H \mathbf{a}(\theta)}, \quad (3)$$

producing a range-angle map $P_{\text{MUSIC}}(\theta, r)$.

3) *Non-Maximum Suppression (NMS)*: To suppress redundant sidelobe peaks, we apply NMS to retain only local maxima above their neighborhood:

$$P_{\text{NMS}}(i, j) = \begin{cases} P(i, j), & \text{if } P(i, j) > P(i', j') \\ & \text{for all } (i', j') \in \mathcal{N}(i, j) \\ 0, & \text{otherwise} \end{cases}$$

4) *Candidate Selection*: Given an expected number of human targets $N = 2$, we select the top $2N$ peaks from the NMS-processed Range-Angle spectrum to improve robustness against weak responses. The NMS window is set appropriately so that each target—typically represented by the chest region—contributes at most one dominant region. In this example, one subject is seated on the left and the other on the right, making them

spatially separable in the angle domain. This ensures sufficient coverage while avoiding redundant detections. Each selected point p_i is associated with a slow-time signal $s_{p_i}(t)$ for further analysis.

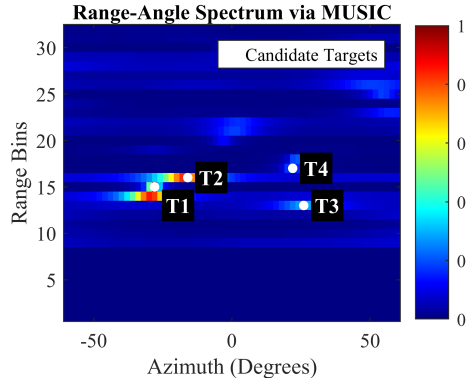


Fig. 2: Candidate target points on the MUSIC-based Range-Angle spectrum. White circles labeled **T1–T4** represent the initial candidate points selected after non-maximum suppression (NMS).

5) *Phase Signal Extraction and Respiration Power Ratio*: For each candidate point p_i , we extract the unwrapped slow-time phase signal:

$$\phi_i(t) = \text{unwrap}(\arg(s_{p_i}(t)))$$

The power spectral density is computed as:

$$\Phi_i(f) = |\mathcal{F}\{\phi_i(t)\}|^2$$

Let $[f_l, f_h] = [0.15, 0.35]$ Hz denote the respiration band, and $f_0 = \arg \max_{f \in [f_l, f_h]} \Phi_i(f)$ the dominant frequency. The respiration Power P_r is defined as:

$$P_r = \Phi_i(f_0 - \Delta f) + \Phi_i(f_0) + \Phi_i(f_0 + \Delta f)$$

with the total spectral power obtained by summing the discrete power spectral density values over the respiration band:

$$P_t = \sum_{f=f_l}^{f_h} \Phi_i(f)$$

The respiration power ratio is:

$$R_i = \frac{P_r}{P_t}$$

We compute R_i for all candidates and retain the top- k points with the highest values for further analysis.

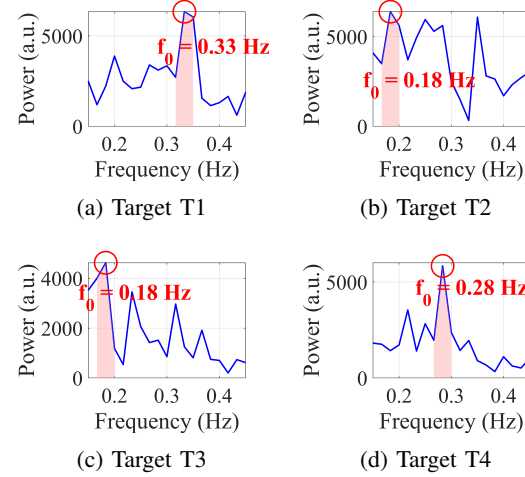


Fig. 3: Power spectra $\Phi_i(f)$ of candidate points T1–T4, showing dominant frequency f_0 and respiration power ratio $R_i = P_r/P_t$. The R_i values are 0.528 (T1), 0.302 (T2), 0.463 (T3), and 0.518 (T4).

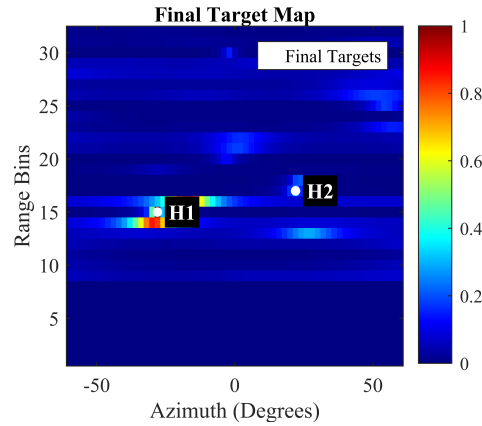


Fig. 4: Final Target Map showing the two selected human targets H1 and H2.

B. Heart Rate Detection

This section presents an improved heartbeat estimation method based on sparse impulse train modeling and frequency-domain analysis. Unlike conventional FFT or autocorrelation approaches

applied to continuous phase signals, our method targets the temporal structure of discrete heartbeat events modeled as sparse impulses.

To improve robustness under low SNR and respiratory interference, we apply a second-order difference followed by CFAR-based spike detection, which suppresses smooth respiratory trends while preserving sharp heartbeat-induced transients. The resulting binary impulse sequence encodes the timing of heartbeat events and is less affected by amplitude noise and low-frequency harmonics.

A bandpass filter and FFT are then applied to the impulse train to estimate the dominant heartbeat frequency. This structure-aware approach offers enhanced stability in noisy or multi-target environments.

1) *CFAR-Based Detection of Heartbeat Events*: Given the slow-time phase signal $\tilde{\phi}[n]$, we apply a second-order difference to enhance heartbeat-related spikes:

$$x[n] = \left| \tilde{\phi}[n+2] - 2\tilde{\phi}[n+1] + \tilde{\phi}[n] \right|$$

CFAR detection [9] is applied to $x[n]$ to identify heartbeat-induced spikes. We adopt a 1D CA-CFAR with 50 training and 12 guard cells on each side (500 ms and 120 ms total), assuming a 100 Hz sampling rate. The noise level is the mean of training cells, scaled by $\alpha = 3.0$. Samples above the threshold are retained as candidate heartbeat events $T = \{t_1, t_2, \dots, t_N\}$.

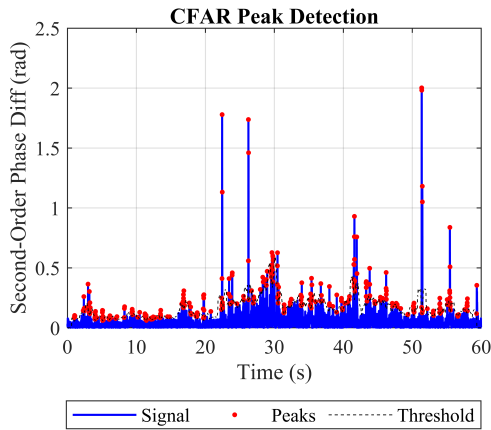


Fig. 5: Detected heartbeat peaks using CFAR on the second-order differenced phase signal.

2) *Impulse Train Construction*: We represent the detected events $\mathcal{T}$ as a discrete impulse train:

$$x[n] = \sum_{i=1}^N \delta(n - t_i)$$

This sparse binary sequence encodes heartbeat timings independent of amplitude, and is zero-padded to a fixed length for spectral analysis.

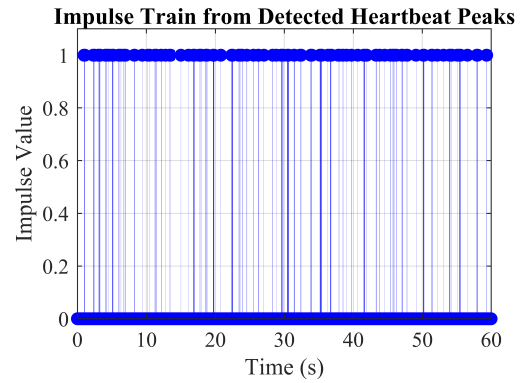


Fig. 6: Impulse train constructed from CFAR-detected heartbeat events.

3) *FFT-Based Heartbeat Frequency Estimation*: A 4th-order Butterworth bandpass filter with passband $[0.8, 2]$ Hz is applied to suppress noise and low-frequency drift:

$$f_{\text{pass}} = [0.8, 2] \text{ Hz}$$

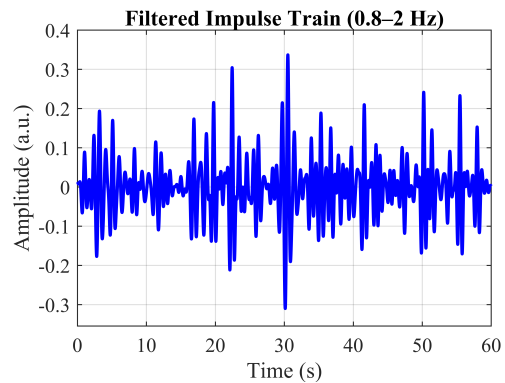


Fig. 7: Filtered impulse train within the heartbeat frequency band.

Let $\hat{x}[n]$ denote the filtered sequence. The power spectrum is computed as:

$$X(f) = |\mathcal{F}\{\hat{x}[n]\}|^2$$

The heartbeat frequency is estimated by locating the dominant peak within the target band:

$$f_0 = \arg \max_{f \in [f_{\min}, f_{\max}]} X(f)$$

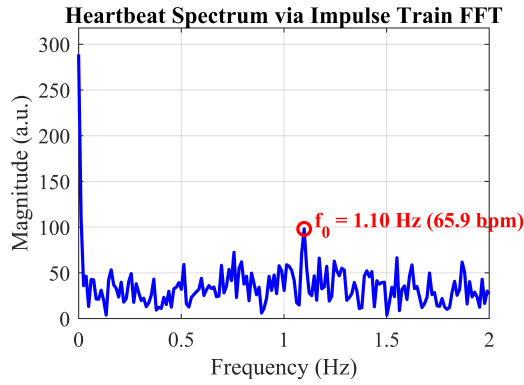


Fig. 8: Heartbeat spectrum estimated from the impulse train. The peak f_0 corresponds to the heartbeat rate.

III. EXPERIMENTS

A. Hardware and Data Collection

The experimental system consists of a Texas Instruments AWR1642 mmWave radar board and a DCA1000EVM data capture card. Raw ADC data was collected using mmWave Studio and stored to a host PC for offline processing in MATLAB. The radar operates at a center frequency of 77 GHz and was mounted on a tripod at chest height. Signal processing procedures—including range FFT, target point selection, phase unwrapping, and vital sign detection—were conducted entirely offline.

B. Parameter Settings

The main radar and signal processing parameters used in the experiments are summarized in Table I. Each frame consisted of a single chirp, and data was collected at a frame rate of 100 Hz.

TABLE I: Radar and Processing Parameter Settings

Parameter	Value
Radar frequency	77 GHz
Frequency slope	60MHz/ μ s
Bandwidth	3600 MHz
Range resolution	4.17 cm
Angle Step Size	2°
Number of ADC samples	256
Range FFT size	256
Frame rate	100 Hz
Frame duration	10 ms
CFAR training cells	50 (500 ms window)
CFAR guard cells	12 (120 ms window)
CFAR threshold factor	3.0
Breathing frequency band	0.15–0.35 Hz
Heartbeat frequency band	0.8–2.0 Hz
NMS window size	5 × 5

C. Experimental Scenarios

Six experiments were conducted with four human subjects, including two two-person and four single-person scenarios. Each 60-second session was performed under varying distances (0.8 m and 1.2 m) and angles (0° and 30°), during which radar signals capturing chest micro-motions were recorded. Ground truth heart rates were measured using an Apple Watch Series 9, which provides reliable PPG-based monitoring and is widely used in consumer-grade physiological tracking, and all spatial parameters were manually calibrated. The primary goal was to detect the subject's heartbeat.



Fig. 9: Experimental setup: The mmWave radar module (AWR1642) and data acquisition board (DCA1000) are positioned in front of a human target. The PC is used to control the radar and collect data for further signal processing.

D. Heartbeat Detection Results

TABLE II: Heartbeat Detection Results

Experiment Setup				Detection Results			
Exp	Subj	Dist	Angle	Ref	FFT	ACF	Ours
1	A	0.8	30	71	63	67	69
1	B	0.8	-30	67	60	61	64
2	C	1.2	30	72	81	78	71
2	D	1.2	-30	75	79	69	71
3	A	1.2	0	73	77	66	70
4	B	1.2	30	70	62	65	68
5	C	0.8	0	71	64	74	72
6	D	0.8	30	76	81	84	73

Note: Dist in meters; Angle in degrees; all heart rate values in beats per minute (bpm).

As shown in Table II, our method achieves the highest accuracy with an average error of 3.29%, significantly outperforming FFT (9.11%) and ACF (7.80%). The results demonstrate better stability and closer estimates to the reference heart rates across different subjects and conditions.

IV. CONCLUSION

This paper presented a non-contact heartbeat detection method based on a mmWave radar system. The proposed approach integrates range-angle spectral estimation, respiration-based target point selection, and a frequency-domain heartbeat detection algorithm using impulse train modeling. By analyzing chest micro-motion signals under varying distances and angles, the method demonstrated improved robustness and accuracy. Experiments involving multiple human subjects showed that our approach significantly reduced the average detection error to 3.29%, compared to 9.11% and 7.80% for conventional FFT and ACF methods, respectively.

These results validate the effectiveness and robustness of our proposed framework for non-contact heartbeat detection using mmWave radar. The system achieves consistently low detection errors across different subjects and geometric configurations, demonstrating its suitability for practical deployment in indoor healthcare and monitoring applications.

For future work, we aim to further improve performance in multi-target and dynamic environments, where signal interference, occlusion,

or subject movement can present additional challenges. In particular, we plan to incorporate adaptive spatial filtering and motion compensation techniques to enhance signal extraction under these conditions. Additionally, we will extend the system to support simultaneous real-time monitoring of both respiration and heartbeat, enabling more comprehensive physiological tracking for use in applications such as sleep monitoring, elderly care, and human-computer interaction. Furthermore, we will explore methods to reduce the required measurement duration while maintaining detection accuracy, in order to support faster assessments and improve user experience in real-time settings.

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